

NEURO LOCKDOWN

10:47 PM. Tomorrow is Maya's big science test. Your class has been shrunk to the size of a single cell and sent into her brain to help her study. The moment you arrive, everything goes into LOCKDOWN. Unlock the **Brain Box** before morning.

How it works

- **Six paper stations** around the room, one per brain region. Each has field notes (the science) and a hands-on puzzle. Solving it gives the code to one lock on the Brain Box.
- **One Brain Terminal**, a laptop or Chromebook running the game in Classroom mode. It shows the countdown and holds the two rooms that need sound and timing: the **Amygdala** and the **Temporal lobes**. The terminal gives the code for the Synapse Lab pouch.
- **The Brain Box** has six locks. Inside is the **Frontal lobe** finale: letter tiles to unscramble, a very tempting red envelope, and the final override, typed into the terminal.

At a glance

Players	6-10 per kit (the whole team works at once, splitting across stations)
Time	30 or 45 minutes, plus 5 min briefing and 5 min debrief
Tech	1 laptop/Chromebook with sound. Open index.html , choose <i>Classroom</i> .
Covers	All 12 primary and 6 supporting terms from the project
Cost	About \$35-50 for locks (reusable), or \$0 with the envelope option

The flow

Stations 1-5

Paper puzzles, any order, all at once. Each opens one lock on the Brain Box.

Brain Terminal

Calm the Amygdala → tune the Sound Lab → get the Synapse pouch code.

Station 6: Synapse

Materials are in the locked pouch. Solving it reveals where the Brain Box key is hidden.

Brain Box → Frontal lobe

All six locks open → unscramble the tiles → type the override at the terminal. Escape!

Why hybrid?

Players physically handle the science: they wire yarn through a thalamus, carry a code across the room in their memory, snap neurotransmitters into receptors and argue over which structures belong in the limbic system. The laptop is used only where tech does something paper can't: a live stress alarm that responds to breathing, and a radio you have to tune by ear.

Supplies & setup

Locks and containers

Item	Code	For
Lockable box (tackle box, toolbox, or box + chain)	—	The Brain Box
6-hole lockout hasp (~\$5) or chain	—	Holds the locks
4-digit combination lock	2075	Brainstem
Directional lock	→ ↑ ↓ ←	Thalamus
5-letter word lock	FOCUS	Hypothalamus
4-digit combination lock	3816	Hippocampus
3-digit combination lock	587	Limbic system
Keyed padlock	key	Synapse (key in VESICLE)
Zipper pouch + 3-digit lock	862	Synapse pouch

Tape a **lock tag** (page 14) to each lock so teams know which code goes where.

Craft supplies

- Printer (color is nice, black and white works)
- Scissors, glue stick, tape
- Cardboard or a folder to back the thalamus switchboard
- Yarn or string: 5 pieces, about 24 in each
- 4 small containers (plastic eggs, envelopes, or cups) for the vesicle hunt
- 1 red envelope
- Optional: candy or prizes for the Brain Box

No locks? Envelope option (\$0)

Put everything that would be in the Brain Box into a big envelope at your desk. Teams bring each code to you. When a code is right, give them one "lock" sticker. Six stickers earns the envelope. Do the same with a small envelope for the Synapse pouch.

Setup checklist (about 20 minutes)

1. Print this kit. Pages marked GAME MASTER ONLY stay with you.
2. Cut out all cards along the dashed lines. Fold the Brainstem fuel cards so the answer side is inside (don't glue).
3. Glue the Thalamus switchboard to cardboard. Poke holes at the circles and tie one yarn piece to each sense.
4. Tape the **Memory Vault poster** on a wall far from Station 4.
5. Lay out Stations 1–5 on separate desks with their field notes page and cards. Shuffle each deck.
6. Put the Synapse materials (pages 12–13) in the pouch. Lock it with **862**.
7. Put the key in the container labeled **VESICLE**. Hide it and the 3 decoy containers around the room.
8. Put the Frontal lobe notes, letter tiles, red envelope and prizes in the Brain Box. Lock all six locks.
9. Open **index.html** on the terminal. Choose *Classroom: Brain Terminal + printed kit* and a clock. Don't start yet.
10. Read the briefing (page 3), then press **Enter the brain**. The clock starts.

KEEP THIS PAGE FACE DOWN

Answer key & hint script

Station	Code	Solution	If they're stuck, say...
1 Brainstem	2075	Glucose is the right fuel card. Its back says: code = oxygen % then water %. Oxygen 20, water 75.	"What does the brain actually burn for energy? Check the supply report."
2 Thalamus	→ ↑ ↓ ←	1 Sight → Occipital (right). 2 Touch → Parietal (up). 3 Smell skips the thalamus, no arrow. 4 Sound → Temporal (down). 5 Movement → Frontal (left).	"Is there a sense that never stops at the relay station?"
3 Hypothalamus	FOCUS	Alert 1 low glucose → Hunger F. 2 cold → Shiver O. 3 hot → Sweat C. 4 salty blood → Thirst U. 5 night → Sleepiness S. (Stress hormones T is a decoy.)	"For each alert, what would bring the body back to normal?"
4 Hippocampus	3816	Memory boosters in notes order: Sleep (moon) 3, Practice (quiz) 8, Water (drop) 1, Calm (circles) 6.	"Split the job. Each keeper only has to remember one or two numbers."
5 Limbic	587	Amygdala 128 + Hippocampus 205 + Hypothalamus 74 + Cingulate gyrus 180. Decoys: Brainstem, Cerebellum, Occipital lobe, Spinal cord.	"The field notes name all four team members."
Terminal	862	Amygdala: hold button ~4 s, release ~4 s, three times. Sound Lab: tune until SIGNAL LOCKED, password WERNICKE . Screen shows pouch code 862.	"Slow down. Count to four in, four out." / "Drag the dial slowly and watch the words."
6 Synapse	VESICLE	1 ▲ V · 2 ● E · 3 ■ S · 4 ● I · 5 ▲ C · 6 ▲ L · 7 ■ E. The key is in the container labeled VESICLE.	"Dopamine is 'do it again.' Serotonin is mood. GABA is the brakes."
Frontal lobe	LEARNING	Unscramble the 8 tiles, type it at the terminal. The red envelope is an impulse-control trap.	"What does your brain do at school all day?"

Running the game

- Give **3 free hints** per team. After that, each hint costs 1 minute (subtract it on the board).
- If a team opens the red envelope, add 1 minute. The card inside explains why.
- The terminal's own hints cost 30 seconds and are tracked on screen.
- Don't give answers. Point back to the field notes. Every answer is in them.

Debrief questions (5 min)

1. Which station made you use your frontal lobe the most? Why?
2. When the clock got low, what did your amygdala do? What helped?
3. What memory strategy did your team use at the Memory Vault?
4. Name one habit from today that would help your brain before a test.

Briefing (read aloud when you're ready to start)

"It's 10:47 PM, and Maya has a science test tomorrow. You've been shrunk down and sent into her brain to help her study. But the second you arrived, her brain went into lockdown. Everything you need is in this room: six stations, each one a part of the brain, and a terminal that's the only computer still running. Every station has field notes. Read them. The answers are always in the science. Solve a station and you'll get a code for one lock on the Brain Box. Open all six, and you can override the lockdown. Your time starts... now."



BRAINSTEM

Life Support

OPENS
4-digit lock

tagged BRAINSTEM

Alarms everywhere. Maya's breathing and heartbeat are stuttering. The brain's autopilot needs the right supplies, and fast.

The **brainstem** is the oldest part of your brain. It sits at the base of your skull and connects your brain to your spinal cord.

WHAT IT RUNS, AUTOMATICALLY

- Breathing
- Heartbeat and blood pressure
- Swallowing, coughing and sneezing
- Waking up and staying alert

It has three parts: the midbrain, the pons and the medulla. You never have to remember to breathe, because the brainstem does it for you, even while you sleep.

SUPPLY REPORT

Oxygen

Your brain is only about **2%** of your body weight, but it uses about **20%** of the **oxygen** you breathe.

Glucose

Glucose is a simple sugar from food, carried in your blood. It is the brain's **main fuel**. Neurons can't store much of it, so they need a steady supply. That's why skipping breakfast can make it hard to focus.

Water

Brain tissue is about **75% water**. Even mild dehydration can make it harder to pay attention and remember.

YOUR MISSION

Step 1. Four fuel cards are on this desk. Only one can power a brain. Pick the one you think is right and open it. **Step 2.** The correct card tells you how to build the code from the supply report.

OXYGEN LINE

— — %

WATER LEVEL

— — %

FUEL LINE

?

× STATION 1 • FUEL CARDS • cut out each card, fold on the middle line so the answer is inside (don't glue)

FUEL CARD • FRONT

PROTEIN

from meat, beans, eggs

INSIDE

FLATLINE

Protein builds and repairs your body, but it isn't the brain's everyday fuel. Try another card.

FUEL CARD • FRONT

CAFFEINE

from soda, coffee, energy drinks

INSIDE

FLATLINE

Caffeine can make you feel alert, but it has no energy for neurons. Too much can even hurt sleep. Try another card.

FUEL CARD • FRONT

GLUCOSE

from food, carried in blood

INSIDE

HEARTBEAT DETECTED

Correct fuel! Now finish the repair. The lock code is the **oxygen %** followed by the **water %** from the supply report.

FUEL CARD • FRONT

SALT

from chips, fries, pretzels

INSIDE

FLATLINE

Your nerves need a little salt to send signals, but it's not fuel. Too much just makes you thirsty. Try another card.



THALAMUS

The Relay Station

OPENS

Directional lock

tagged THALAMUS

Signals from Maya's senses are piling up in the middle of her brain with nowhere to go. Somebody has to run the switchboard.

The **thalamus** is a pair of egg-shaped structures right in the middle of your brain. It is the brain's **relay station**, like a switchboard operator.

- Almost every signal from your senses stops at the thalamus first. It sorts the signal and sends it on to the right part of the brain's outer layer, the cortex.
- It also relays movement signals and helps control how awake and alert you feel.

WHERE SIGNALS GO

Occipital lobe

At the back of your head. Handles **vision**.

Parietal lobe

On top, behind the frontal lobe. Handles **touch**, pain, temperature and body position.

Temporal lobes

On the sides, by your ears. Handle **hearing**.

Frontal lobe

At the front. Its motor cortex turns **movement** plans into action.

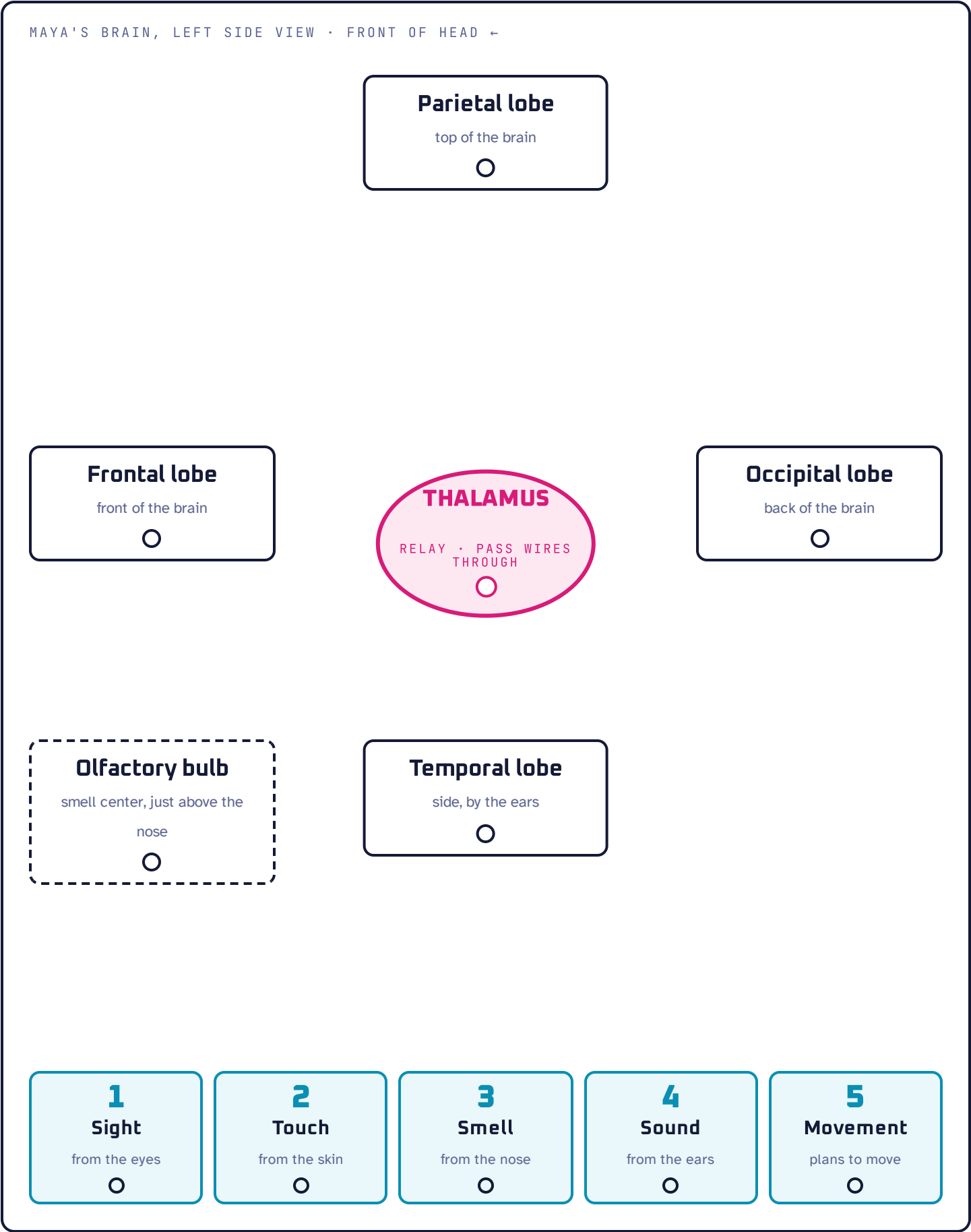
THE ONE EXCEPTION

Smell skips the thalamus. Smell signals travel from your nose to the **olfactory bulb**, then straight to areas wired to the amygdala and hippocampus. That's why a smell can bring back a memory or feeling instantly.

YOUR MISSION

On the switchboard, run each sense's yarn **through the thalamus hole** in the middle, then to the part of the brain that handles it.

Your code is the **direction each wire leaves the thalamus** (→ ↑ ↓ ←), in sense-number order, 1 to 5. A signal that doesn't pass through the thalamus adds nothing to the code.





HYPOTHALAMUS

The Thermostat

OPENS

5-letter word lock

tagged HYPOTHALAMUS

Every body system is out of balance, and the alerts keep coming. The control tower needs an operator.

The **hypothalamus** is about the size of an almond and sits just below the thalamus. *Hypo* means "below." It is your body's **thermostat and control tower**.

Its big job is **homeostasis**: keeping your body balanced, no matter what's happening outside.

HOW IT RESTORES BALANCE

Temperature

Too hot? It makes you **sweat** so you cool off. Too cold? It makes you **shiver** so your muscles make heat.

Water

When you're low on **water** and your blood gets too salty, it makes you feel **thirsty**.

Food

When blood **glucose** drops, it switches on **hunger**.

Sleep

Its built-in body clock makes you **sleepy** when it gets dark at night.

Stress

When the amygdala senses danger, the hypothalamus signals glands to release **stress hormones** like cortisol and adrenaline.

YOUR MISSION

Lay the five alert cards in number order. Under each one, place the response the hypothalamus would trigger to fix it. Read the letters on your responses, in order, to open the lock.

× ALERT CARDS

ALERT 1 Blood glucose dropping. Last meal was 6 hours ago.	ALERT 2 Body temp 35.9 °C. Waiting for the bus in the snow.	ALERT 3 Body temp 38.4 °C and climbing after gym class.	ALERT 4 Water level low. Blood is getting too salty.	ALERT 5 10:30 PM. Dark room. Body clock says: night.
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× RESPONSE CARDS

Sweat C	Stress hormones T	Thirst U	Hunger F	Sleepiness S	Shiver O
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HIPPOCAMPUS

The Memory Vault

OPENS

4-digit lock

tagged HIPPOCAMPUS

Tonight's studying is about to be lost. The code is stored in the Memory Vault on the far wall, and you can only carry it back in your head.

Hippocampus is Greek for **seahorse**, because it's curled like one. You have two, one inside each temporal lobe.

- It turns **short-term memories into long-term memories**.
- It helps you remember facts and events from your life.
- It helps you find your way, like remembering the route to a new classroom.

MEMORY BOOSTERS

1. Sleep

While you sleep, the hippocampus replays what you learned that day and helps store it for good.

2. Practice

Quizzing yourself builds stronger **memory** than rereading notes.

3. Water

A hydrated, well-fed brain remembers better.

4. Calm

Stress that lasts a long time floods the brain with cortisol, which can harm the hippocampus.

MEMORY TRICKS

Chunking: group numbers into pairs, like 38-16.

Rehearsal: repeat it out loud all the way back.

Teamwork: each person remembers one piece.

YOUR MISSION

Find the Memory Vault poster somewhere in the room. The code is the numbers next to the **four memory boosters** from these notes, in the order they're listed.

Vault rules: No paper, pencils, phones or photos at the vault. Only **one** person at the vault at a time. Carry the numbers back in your memory. Switch runners as often as you like.

THE MEMORY VAULT

Look. Remember. Walk away. No writing, no phones.



3

sleep



5

phone



8

practice quiz



2

soda



9

video game



1

water



4

alarm clock



7

candy



0

headphones



6

calm breathing



5

TV



2

pizza



LIMBIC SYSTEM

Emotion Headquarters

OPENS

3-digit lock

tagged LIMBIC

Maya's emotion and memory team has been scattered all over the brain. Reassemble the team, and only the team.

The **limbic system** is a team of structures deep in the middle of the brain that work together on **emotion, memory and motivation**.

Limbus is Latin for "edge" or "border." The team forms a curved ring around the brain's core.

THE TEAM

Amygdala

Emotions, especially fear. The brain's alarm.

Hippocampus

Forms new memories.

Hypothalamus

Connects emotions to the body: heart rate, hormones, hunger.

Cingulate gyrus

A curved belt of cortex above the brain's center. Helps you notice and manage emotions and keep your attention on a task.

Many scientists also count the thalamus as a partner of the team.

WHY IT MATTERS AT SCHOOL

Emotion and memory are wired together. You remember things better when you care about them, which is why a fun or surprising lesson sticks.

YOUR MISSION

Eight structure cards are on this desk. Only four are on the limbic team. Find them, then **add up their numbers**. The total opens the lock.

< STRUCTURE CARDS

Brainstem

base of the skull

143

Amygdala

almond-shaped

128

Cerebellum

back, below the cerebrum

91

Hypothalamus

below the thalamus

74

Occipital lobe

back of the head

212

Cingulate gyrus

curved belt of cortex

180

Spinal cord

runs down the back

57

Hippocampus

seahorse-shaped

205



NEUROTRANSMITTERS

The Synapse Lab

REVEALS

Where the key is
for the KEY Lock

Maya's neurons have stopped talking to each other. Load the right chemical messenger into each synapse.

Your brain has about **86 billion neurons**. They talk using electricity and chemistry.

- A signal zips down a neuron as an electrical pulse.
- At the end it reaches a tiny gap called a **synapse**.
- Tiny bubbles called **vesicles** release **neurotransmitters**, chemical messengers that cross the gap and fit into **receptors** on the next neuron, like a key in a lock.

THE MESSENGERS

▲ Dopamine

Reward and motivation. Released when you reach a goal or expect something good. Makes you want to do it again. Also helps control smooth movement.

● Serotonin

Mood, sleep and appetite. Helps you feel steady and calm. About 90% of your body's serotonin is made in your gut!

■ GABA

The brain's brakes. Tells neurons to slow down and fire less, which helps you feel calm and fall asleep.

YOUR MISSION

Match each of the 7 situation cards with the messenger that fits by placing a molecule token on it. Then use the decoder: for each card number, find the letter under that molecule's shape. The 7 letters name the place where Maya's key is stored. Find it in the room.

Situation	▲ Dopamine	● Serotonin	■ GABA
Card 1	V	K	T
Card 2	O	E	A
Card 3	R	N	S
Card 4	U	I	M
Card 5	C	P	D
Card 6	L	B	H
Card 7	Y	W	E

× STATION 6 • SITUATION CARDS (put in the pouch)

SITUATION CARD 1

Maya finally beats a level she's been stuck on for days, and she can't wait to play again.

place a molecule token here

SITUATION CARD 2

A good night's sleep and morning sunlight help Maya's mood stay steady all day.

place a molecule token here

SITUATION CARD 3

Maya's racing thoughts slow down so she can finally fall asleep.

place a molecule token here

SITUATION CARD 4

This messenger is mostly made in the gut and helps control appetite.

place a molecule token here

SITUATION CARD 5

Maya checks off her first homework task and feels motivated to start the next one.

place a molecule token here

SITUATION CARD 6

This messenger helps Maya's muscles move smoothly when she shoots a basketball.

place a molecule token here

SITUATION CARD 7

The brakes: this messenger tells overexcited neurons to fire less.

place a molecule token here

HOW NEURONS TALK

Electrical pulse → vesicles release neurotransmitters → they cross the synapse → they fit into receptors on the next neuron.

× MOLECULE TOKENS • 3 of each



Dopamine



Dopamine



Dopamine



Serotonin



Serotonin



Serotonin



GABA



GABA

× CONTAINER LABELS • put the key in VESICLE, hide all four around the room

VESICLE

stores neurotransmitters until the signal arrives

AXON

the long wire a signal travels down

DENDRITE

branches that receive signals

MYELIN

fatty coating that speeds signals up

× POUCH & BOX LABELS

SYNAPSE LAB

Locked. The code is on the Brain Terminal.

THE BRAIN BOX

Six locks. Six regions. One way out.

× LOCK TAGS • tape one to each lock

BRAINSTEM

4-digit lock

THALAMUS

directional lock

HYPOTHALAMUS

word lock

HIPPOCAMPUS

4-digit lock

LIMBIC

3-digit lock

SYNAPSE KEY

key lock

SYNAPSE POUCH

3-digit lock

EXTRA

spare lock

× BRAIN TERMINAL SIGN • stand it next to the laptop

BRAIN TERMINAL

The only computer still running in Maya's brain. Two rooms are inside: the **Amygdala** and the **Temporal lobes**. Turn the sound up. When you open the Brain Box, come back here for the final override.



FRONTAL LOBE

Command Center · inside the Brain Box

FINAL

Override code

typed at the terminal

The **frontal lobe** sits right behind your forehead. It's the largest lobe and the brain's **command center**.

- Planning, organizing and making decisions
- Focus and solving problems
- **Impulse control**: stopping to think before you act
- Personality and behavior
- Movement (the motor cortex) and speaking (Broca's area)

STILL UNDER CONSTRUCTION

The very front part, the **prefrontal cortex**, keeps developing until about age 25. Every time you practice planning, focusing or pausing before you react, you strengthen those connections.

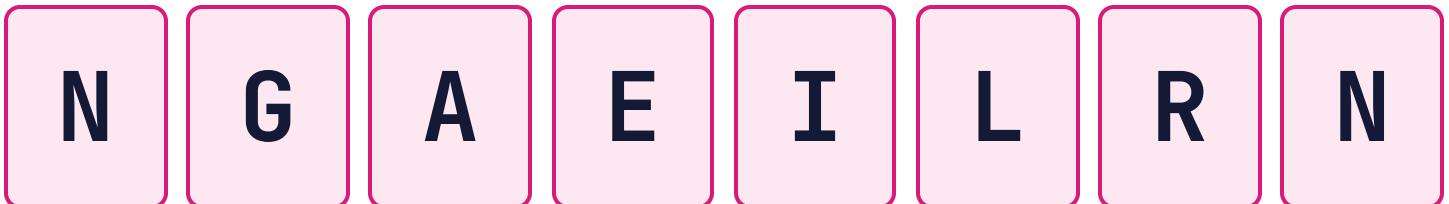
CONNECTIONS

When the amygdala's alarm is too loud, the frontal lobe struggles. Mindfulness, sleep, water and glucose all help it work at full power.

FINAL MISSION

Arrange the eight letter tiles into the override code. Then type it into the Brain Terminal's **Frontal lobe** room. Plan first. Don't guess.

× LETTER TILES · shuffle before putting in the box



× PUT IN THE RED ENVELOPE. WRITE ON IT:
"INSTANT ESCAPE: THE ANSWER IS INSIDE"

IMPULSE DETECTED

That envelope was a test of your **frontal lobe**. Its job is **impulse control**: stopping to think before you act. There is no instant escape, only planning.

Take this card to the Game Master. +1 minute.

× PUT IN THE BOX

You opened Maya's brain.

Every lock was a part of the brain working together. One step left: the command center. Head to the Brain Terminal.



NEURO LOG

One per team · fill it in as you go

TEAM _____

Scientists record what they find. Fill in each blank from the field notes at that station. Your Game Master may check this before you get a hint.

- 1 Brainstem** The brain is 2% of body weight but uses _____% of the oxygen. Its main fuel is _____.
- 2 Thalamus** The thalamus is the brain's _____ station. The one sense that skips it: _____.
- 3 Hypothalamus** Keeping the body balanced is called _____. When you're too hot, it makes you _____.
- 4 Hippocampus** It turns short-term memories into _____ memories. Our team's memory trick: _____.
- 5 Limbic system** This team works on emotion, _____ and motivation. Name two members: _____, _____.
- Terminal: Amygdala** The amygdala is the brain's _____ system. One way to calm **stress** using **mindfulness**: _____.
- Terminal: Temporal** The temporal lobes handle _____ and understanding _____. They sit just above your _____.
- 6 Synapse** Neurotransmitters cross a gap called the _____. Dopamine = _____, serotonin = _____, GABA = _____.
- Frontal lobe** The frontal lobe controls planning, decisions and _____ control. It finishes developing around age _____.

Lock tracker

Brainstem

Thalamus

Hypothal.

Hippoc.

Limbic

Pouch

Key in...
